

Week 2

2.2 | Mapping machine learning tasks



 30 minutes

Machine learning is a subfield of artificial intelligence (AI) that focuses on developing algorithms and models that enable computers to learn from data and make predictions or decisions without explicit programming. In essence, it's about creating systems that can improve their performance over time by recognising patterns and gaining insights from large volumes of data. At its core, machine learning relies on statistical techniques and computational methods to teach computers how to generalise from past experiences and adapt to new information. It involves training models on historical data, allowing them to identify trends, relationships, and anomalies, which can then be applied to make predictions or automate decision-making processes.

In the business world, machine learning has found a wide range of applications that are revolutionising how companies operate and make strategic decisions. One of the most prominent areas of application is in predictive analytics. Businesses use machine learning to forecast future trends, customer behaviour, and market demand, helping them optimise inventory, manage supply chains efficiently, and make informed marketing decisions. Customer relationship management (CRM) has also been greatly enhanced by machine learning, as it enables companies to personalise customer experiences, recommend products, and target marketing campaigns more effectively. Moreover, fraud detection and cybersecurity have become more robust with machine learning algorithms that can identify unusual patterns and detect potential threats in real time. Machine learning also plays a crucial role in natural language processing, enabling chatbots and virtual assistants to provide customer support and automate routine interactions.

The challenge

You will examine several business scenarios and conduct an exercise to map business goals with machine learning technologies and related data so you can build your understanding of key concepts and applications of machine learning.

The output



Completed mapping task of business scenarios, data, and machine learning types (regression, classification, clustering, and anomaly detection).

The task

In this task, you will watch two videos and read a short article to learn about basic machine learning types and their applications. You will then review a set of business scenarios and identify the most appropriate machine learning type and explain the role of data in the scenario.

Note: Accessing the LinkedIn Learning video content through your Leganto reading list will provide you with free access to this LinkedIn Learning content.

Instructions

Complete the following steps:

1. Go to the **Reading List** in the navigation bar on the left side of your screen to access and watch the following videos:
 - What is machine learning? (Melo, 2022)
 - Practice: Types of machine learning (Melo, 2022)
2. Read the following article:
 - [Machine Learning Algorithm Cheat Sheet for Azure Machine Learning designer](#)  (Microsoft Learn, 2024)
3. Complete the following business scenario exercise and review the suggested solutions. Note any questions you may have about the solutions to discuss in this week's live class.

Your progress was restored [Start over](#)

Summary

5 / 5

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BUSM2725_W2C2

▶ Scenario 1 ⓘ

▶ Scenario 2 ⓘ

▶ Scenario 3 ⓘ

▶ Scenario 4 ⓘ

Summary & submit

Total score

4 / 8

4 of 8 interactions

Book progress

0%

0 of 4 pages

Interactions progress

50%

4 of 8 interactions

Submit Report

Restart

Summary

All interactions ▾

Scenario 1 ⓘ

1 of 2 interactions completed

Score

Select the most appropriate machine learning type: 1 / 1

Describe the role of data in this scenario: 0 / 1

Scenario 2 ⓘ

1 of 2 interactions completed

Score

Select the most appropriate machine learning type: 1 / 1

Describe the role of data in this scenario: 0 / 1

Scenario 3 ⓘ

1 of 2 interactions completed

Score

Select the most appropriate machine learning type: 1 / 1

Describe the role of data in this scenario: 0 / 1

https://rmit.instructure.com/courses/170278/pages/2-dot-2-%7C-mapping-machine-learning-tasks?module_item_id=8384037

3/5

Scenario 4

1 of 2 interactions completed



Score

- ☒ Select the most appropriate machine learning type: 1 / 1
- ☐ Describe the role of data in this scenario: 0 / 1

★ 4/8



Summary

5 / 5



Reuse

I am confused



Come prepared to discuss the key concepts in this week's live class.

Watch: Assessment Case Study

Week 2 - 2.2 - Modern Furniture Co. Case Study-Fixed

9:31

If you have yet check the Assessment 1, it is a good time to do so. Please check the simulator and the guideline carefully. Start thinking about what specific tasks you would like to address with the data from this case study.

Supporting resources

- [What is Azure Machine Learning?](#)  (Microsoft Learn, 2024).



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